

DSA0502 – Query Processing for Data Science	
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LAB PROGRAMS 25 AND 26

25. Plot Two or More Lines with Legends, Different Widths and Colors

Question

Write a Python program to plot two or more lines with legends, different widths and colors.

Aim

To plot multiple line graphs with different colors, line widths, and legends.

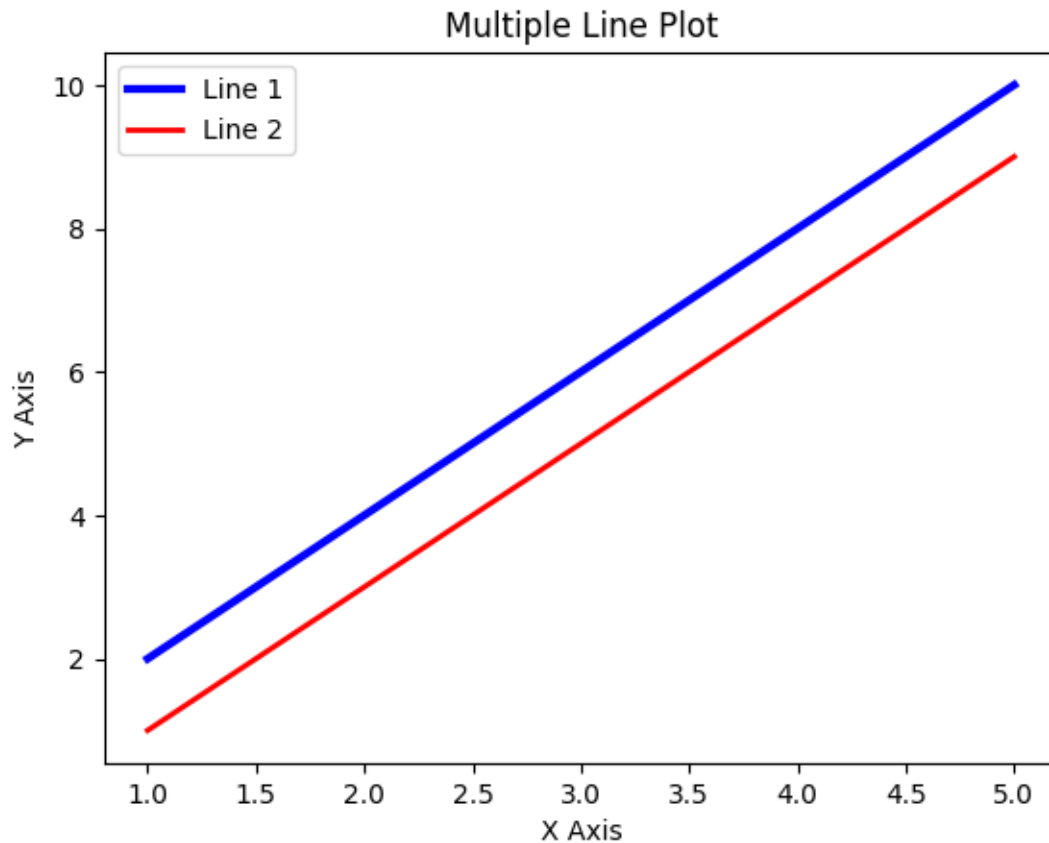
Code

```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y1 = [2, 4, 6, 8, 10]
y2 = [1, 3, 5, 7, 9]
plt.plot(x, y1, color="blue", linewidth=3, label="Line 1")
plt.plot(x, y2, color="red", linewidth=2, label="Line 2")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Multiple Line Plot")
plt.legend()
plt.show()
```

Output

A graph containing:

- Blue Line – Width 3
- Red Line – Width 2
- Legend showing **Line 1** and **Line 2**



Result

Multiple line plots with different colors, line widths, and legends were plotted successfully using Matplotlib.

26. Create Multiple Plots

Question

Write a Python program to create multiple plots.

Aim

To create multiple plots in a single figure using subplot functionality.

Code

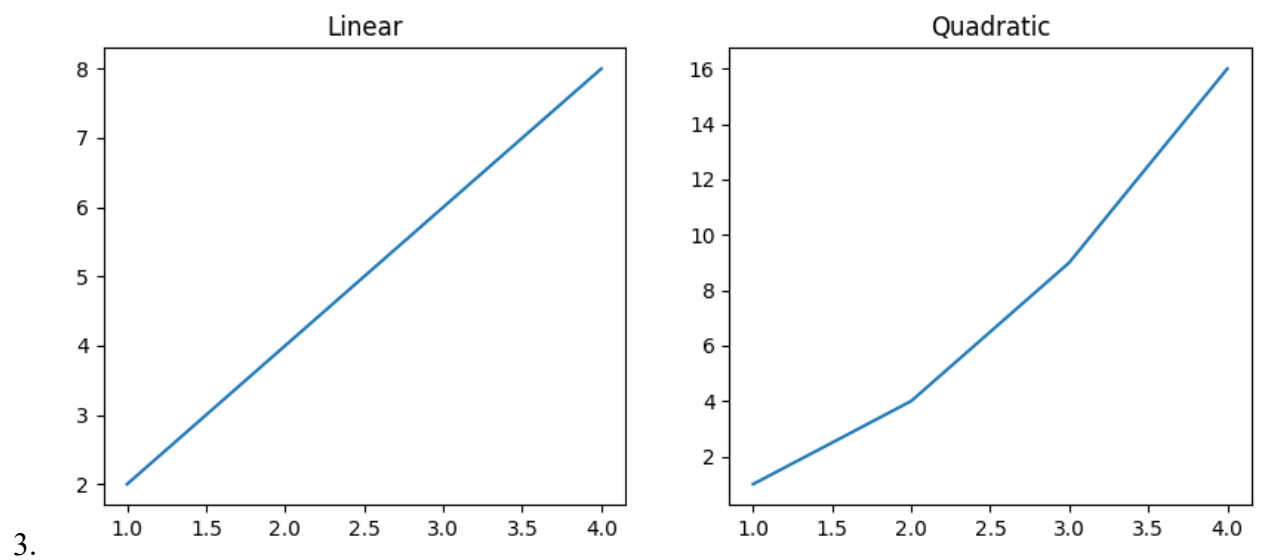
```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4]
y1 = [2, 4, 6, 8]
y2 = [1, 4, 9, 16]
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.plot(x, y1)
```

```
plt.title("Linear")
plt.subplot(1, 2, 2)
plt.plot(x, y2)
plt.title("Quadratic")
plt.show()
```

Output

The output displays two plots side by side in the same figure:

1. **Linear Plot**
2. **Quadratic Plot**



Result

Multiple plots were created successfully using Matplotlib's subplot functionality.